Coordinate frames & tf2

Describe the same point. Move between frames.

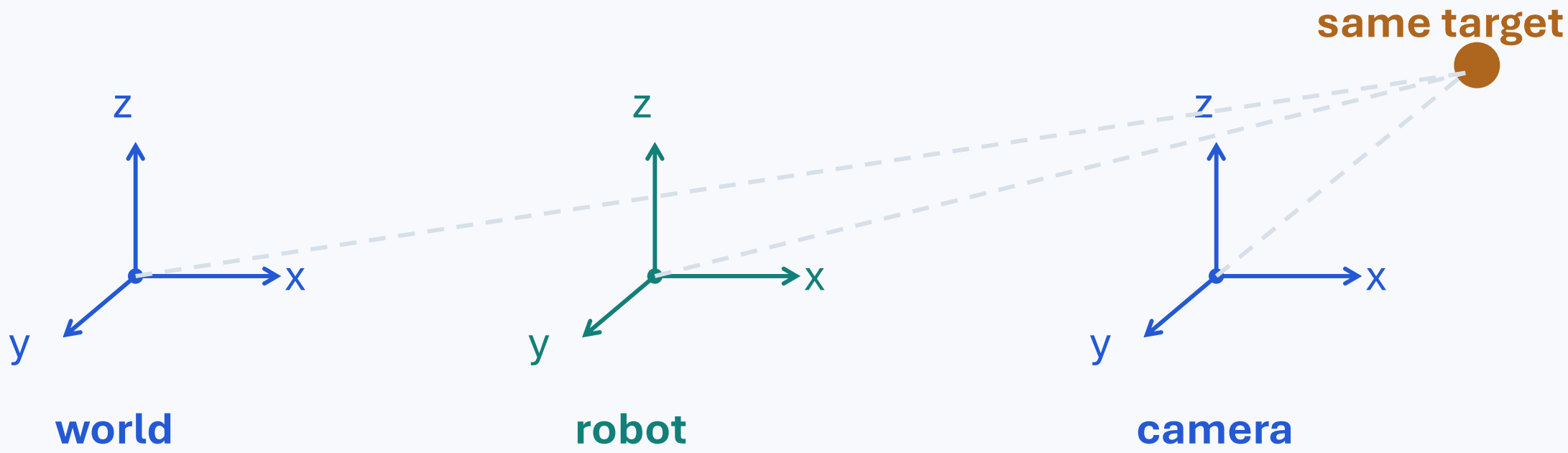
Understand frames | **Read the math** | **Inspect the demo**

$$p_A = R_{AB}\, p_B + t_{AB}$$

20 min concepts · 25 min code, demo & questions

Why do coordinates need a frame?

One physical target. Several coordinate descriptions.



A camera measurement needs a robot-frame destination.

What are TF and tf2?

Geometry describes the relationship; tf2 manages it over time.

Frame

Origin + axes
Where are coordinates measured?

Transform

Rotate + translate
Between frames.

tf2

Time-aware library
Find and apply the relationship.

Always check the conventions

Read the standard and the API before combining data.

ROS body

x forward · y left · z up

Optical

x right · y down · z forward

- ✓ Right-handed · m/rad
- ✓ Transform direction
- ✓ Euler sequence/axes
- ✓ Column/row vectors

Math q: (w, x, y, z)

ROS q: (x, y, z, w)

Read the direction correctly

The point stays still; its coordinate tuple changes.

$$ {}^{A}p = {}^{A}R_{B}\,{}^{B}p + {}^{A}t_{B} $$

B coordinates —${}^{A}T_{B}$→ **A coordinates**

Pose of B in A · maps B → A · same physical point

Matrix representation

Rotation preserves geometry. Translation moves the origin.

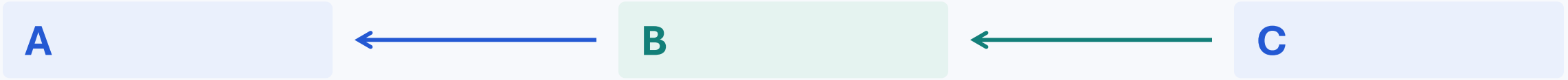
$$R_x(\varphi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c_\varphi & -s_\varphi \\ 0 & s_\varphi & c_\varphi \end{bmatrix} \quad R_y(\theta) = \begin{bmatrix} c_\theta & 0 & s_\theta \\ 0 & 1 & 0 \\ -s_\theta & 0 & c_\theta \end{bmatrix} \quad R_z(\psi) = \begin{bmatrix} c_\psi & -s_\psi & 0 \\ s_\psi & c_\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$T = \begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix} \qquad R^T R = I \qquad \det R = +1$$

Point [p; 1] · Direction [d; 0]

Compose and invert

Follow the frame chain. Apply the rightmost transform first.



$$ {}^{A}T_{C} = {}^{A}T_{B}\,{}^{B}T_{C} $$

$$ T^{-1} = \begin{bmatrix} R^{T} & -R^{T}t \\ 0 & 1 \end{bmatrix} $$

Subtract t, then rotate back.

Euler angles: order matters

This lesson uses fixed-axis XYZ roll–pitch–yaw.

Roll ϕ

about x

Pitch θ

about y

Yaw ψ

about z

$$R = R_z(\psi)\, R_y(\theta)\, R_x(\varphi)$$

$90°$ x then z: $e_x \to e_y$ · z then x: $e_x \to e_z$

Quaternion representation

An axis and angle become four constrained numbers.

$$q = (w, v) = (\cos(\alpha/2), k \sin(\alpha/2))$$

k: unit rotation axis · α: right-hand rotation angle

Unit norm

$w^2 + x^2 + y^2 + z^2 = 1$

ROS identity

(x, y, z, w) = (0, 0, 0, 1)

Rotate with a quaternion

Rotate first; add translation for displaced origins.

$$(0, p_{rot}) = q \otimes (0, p) \otimes q^{-1}$$

$$p_A = p_{rot} + t_{AB}$$

Unit inverse = conjugate · Composition: q_AB ⊗ q_BC

One example, three representations

Same orientation. Same translation. Same base point.

Matrix

$$\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Euler

RPY = (0, 0, π/2)
Fixed XYZ

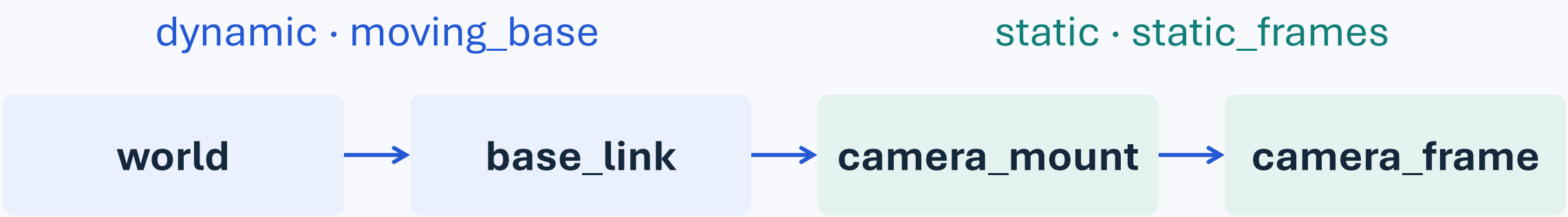
Quaternion

ROS (x,y,z,w):
(0, 0, √2/2, √2/2)

$$(0.2,\ 0,\ 1) \rightarrow (0,\ 0.2,\ 1.5) \qquad t_z = 0.5\ \text{m}$$

The frame tree

Frames are coordinate systems; ROS nodes publish their relationships.

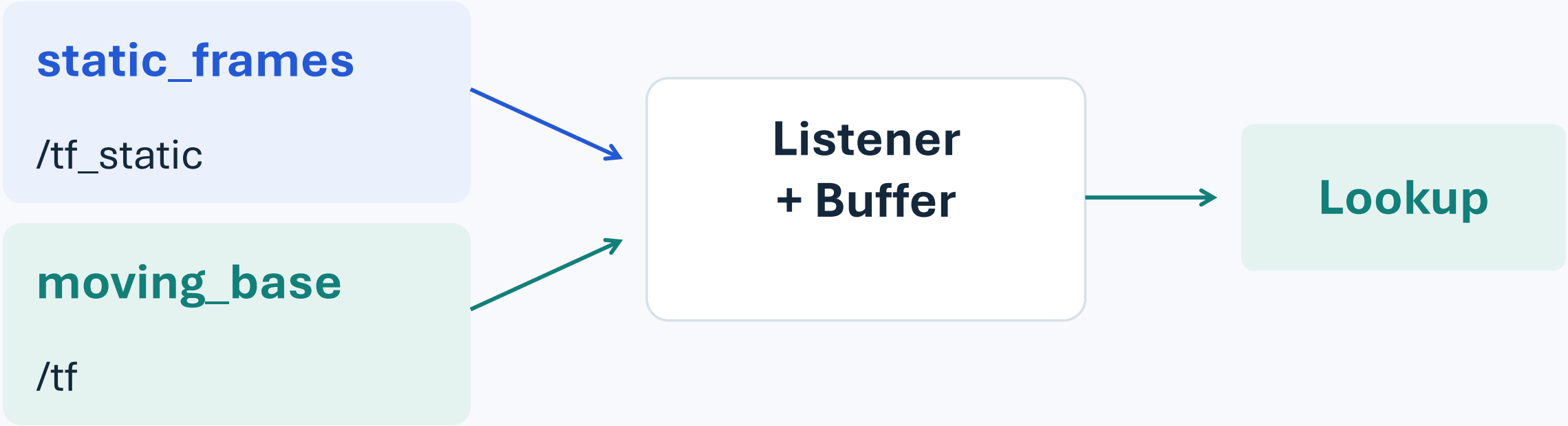


One parent per child · one authority per edge · no cycles

Disconnected trees have no transform path.

How tf2 works

Publish edges. Buffer samples. Look up a connected path.



Several publishers · each listener has its own buffer

Transform at the measurement time

A moving robot makes an old observation different from a new one.



Finite history · consistent clocks · preserve the stamp

Inspect and predict

Different coordinate tuples should overlap in RViz.

```
ros2 run tf2_ros tf2_echo base_link camera_frame

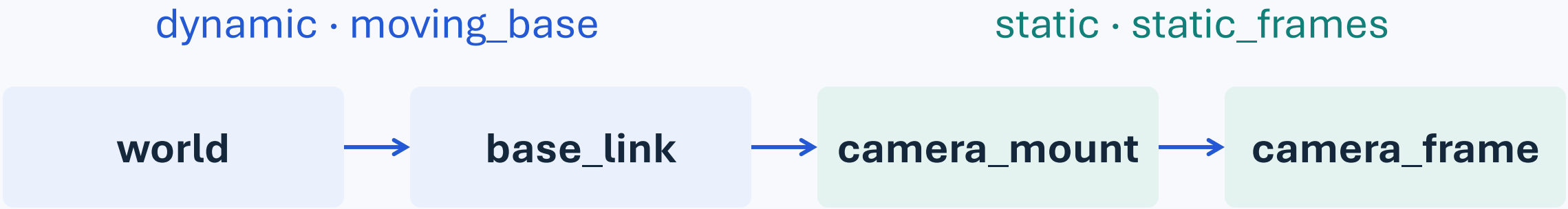
ros2 run tf2_ros tf2_echo world base_link
```

lookup_transform(target, source, time)

RViz Fixed Frame: base_link → world

Live demo: read, run, predict

Keep the geometry and timestamps in view.



Camera (0.2, 0, 1) → Base (0, 0.2, 1.5)

Predict: missing frame? moving base? old timestamp?

OPTIONAL / DEEPER DISCUSSION

Expanded Euler rotation matrix

Fixed XYZ / column vectors: R = Rz(ψ) Ry(θ) Rx(ϕ)

$$\begin{bmatrix} c_\psi c_\theta & c_\psi s_\theta s_\varphi - s_\psi c_\varphi & c_\psi s_\theta c_\varphi + s_\psi s_\varphi \\ s_\psi c_\theta & s_\psi s_\theta s_\varphi + c_\psi c_\varphi & s_\psi s_\theta c_\varphi - c_\psi s_\varphi \\ -s_\theta & c_\theta s_\varphi & c_\theta c_\varphi \end{bmatrix}$$

c = cos(angle) · s = sin(angle) · full angles here

OPTIONAL / DEEPER DISCUSSION

Recover Euler angles

Away from the singularity, recover one principal representative.

$$\theta = \operatorname{atan2}\left(-r_{31}, \sqrt{\left(r_{11}^{\,2} + r_{21}^{\,2}\right)}\right)$$

$$\varphi = \operatorname{atan2}(r_{32}, r_{33}) \qquad \psi = \operatorname{atan2}(r_{21}, r_{11})$$

$\theta \in [-\pi/2, \pi/2]$ · $\phi, \psi \in (-\pi, \pi]$

$$\cos\theta \approx 0: \quad \varphi = 0, \ \psi = \operatorname{atan2}(-r_{12}, r_{22})$$

OPTIONAL / DEEPER DISCUSSION

Quaternion → rotation matrix

Unit Hamilton quaternion q = (w, x, y, z).

$$\begin{bmatrix} 1-2(y^2+z^2) & 2(xy-wz) & 2(xz+wy) \\ 2(xy+wz) & 1-2(x^2+z^2) & 2(yz-wx) \\ 2(xz-wy) & 2(yz+wx) & 1-2(x^2+y^2) \end{bmatrix}$$

Normalize first · R(q) = R(−q) · ROS order is xyzw

OPTIONAL / DEEPER DISCUSSION

Stable matrix → quaternion

Choose the largest component; stay stable near 180°.

$$a_w = (1+r_{11}+r_{22}+r_{33})/4$$

$$a_x = (1+r_{11}-r_{22}-r_{33})/4$$

$$a_y = (1-r_{11}+r_{22}-r_{33})/4$$

$$a_z = (1-r_{11}-r_{22}+r_{33})/4$$

Take √(largest), recover the others, then normalize.

180° about x: (a_w,a_x,a_y,a_z) = (0,1,0,0)

OPTIONAL / DEEPER DISCUSSION

Roll–pitch–yaw → quaternion

Fixed XYZ only; c and s here use HALF angles.

$$w = c_\varphi\, c_\theta\, c_\psi + s_\varphi\, s_\theta\, s_\psi$$

$$x = s_\varphi\, c_\theta\, c_\psi - c_\varphi\, s_\theta\, s_\psi$$

$$y = c_\varphi\, s_\theta\, c_\psi + s_\varphi\, c_\theta\, s_\psi$$

$$z = c_\varphi\, c_\theta\, s_\psi - s_\varphi\, s_\theta\, c_\psi$$

Reverse: normalized q → R(q) → Euler (singular branch)

OPTIONAL / DEEPER DISCUSSION

Quaternion algebra

Hamilton multiplication; order and multiplication side matter.

$$q_1 \otimes q_2 = (w_1 w_2 - v_1 \cdot v_2,\ w_1 v_2 + w_2 v_1 + v_1 \times v_2)$$

$$q^* = (w, -v) \qquad q^{-1} = q^* / \|q\|^2$$

$$q_{rel} = q_2 \otimes q_1^{-1} \qquad R(q) = R(-q)$$

Unit norm: inverse = conjugate. Sign flip ≠ inverse.

OPTIONAL / DEEPER DISCUSSION

Conventions change the formula

Define the operation and vector layout before copying equations.

Active

Rotate the vector: p' = Rp

Passive

Reverse basis map: use R^T

Row vectors: $p'^T = p^T R^T + t^T$

Basis change: $R_{new} = S_A \, R_{old} \, S_B^{-1}$

OPTIONAL / DEEPER DISCUSSION

ROS axes and common frames

Body and optical conventions coexist in one robot.

Body

x forward · y left · z up

Optical

x right · y down · z forward

map → odom → base_link → sensor

map: corrections may jump · odom: continuous, drifts

OPTIONAL / DEEPER DISCUSSION

Interpolate time and orientation

Use compatible buffered samples; choose the short quaternion arc.

$$u = (\tau - \tau_0)/(\tau_1 - \tau_0) \quad t(u) = (1-u)t_0 + ut_1$$

$$q(u) = [\sin((1-u)\Omega)/\sin\Omega]\, q_0 + [\sin(u\Omega)/\sin\Omega]\, q_1$$

$\Omega = \text{acos}(q_0 \cdot q_1)$ · flip q_1 if dot < 0
Near equality: normalize the linear blend.

OPTIONAL / DEEPER DISCUSSION

Worked mathematical checks

Compare physical transforms, not raw angle tuples or quaternion signs.

$$\begin{bmatrix} 0.664463 & -0.733295 & 0.144110 \\ 0.664463 & 0.491450 & -0.562997 \\ 0.342020 & 0.469846 & 0.813798 \end{bmatrix}$$

RPY: (30°, −20°, 45°)
Fixed XYZ

$q \approx (0.861642, 0.299673, -0.057422, 0.405550)$

Round trip · q / −q · near 180° · singular Euler